

Elastic AI Search: Agents, Retrieval, and Observability

Elastic Silicon Valley Meetup - June 23rd 2026.

Agenda

ElasticSeer: AI Search and SRE

Sathvik Vempati (Founder/Developer)

Hybrid Retrieval Patterns for AI Agents

Dan Goosewin (Mastra)

Elastic 2026: The Next Era of Search and Observability

Pramod Saripalli (Principal Customer Architect, Elastic)

Elastic — The Search AI Company

Elastic helps everyone transform data into **answers**, **actions**, and **outcomes** with Search AI.



Founded in
2012



5B+
downloads



3,000+
employees



Used by over **54%**
of the Fortune 500



40+
Countries with employees



Publicly traded under
ESTC in the NYSE



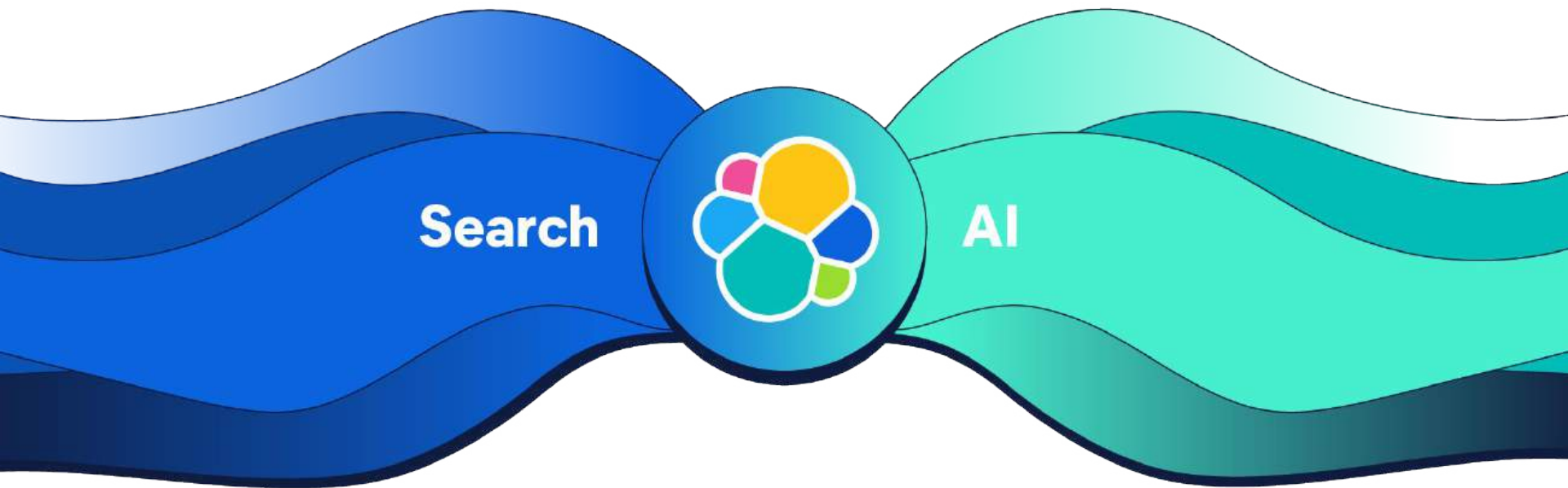
1,500+
Partners globally



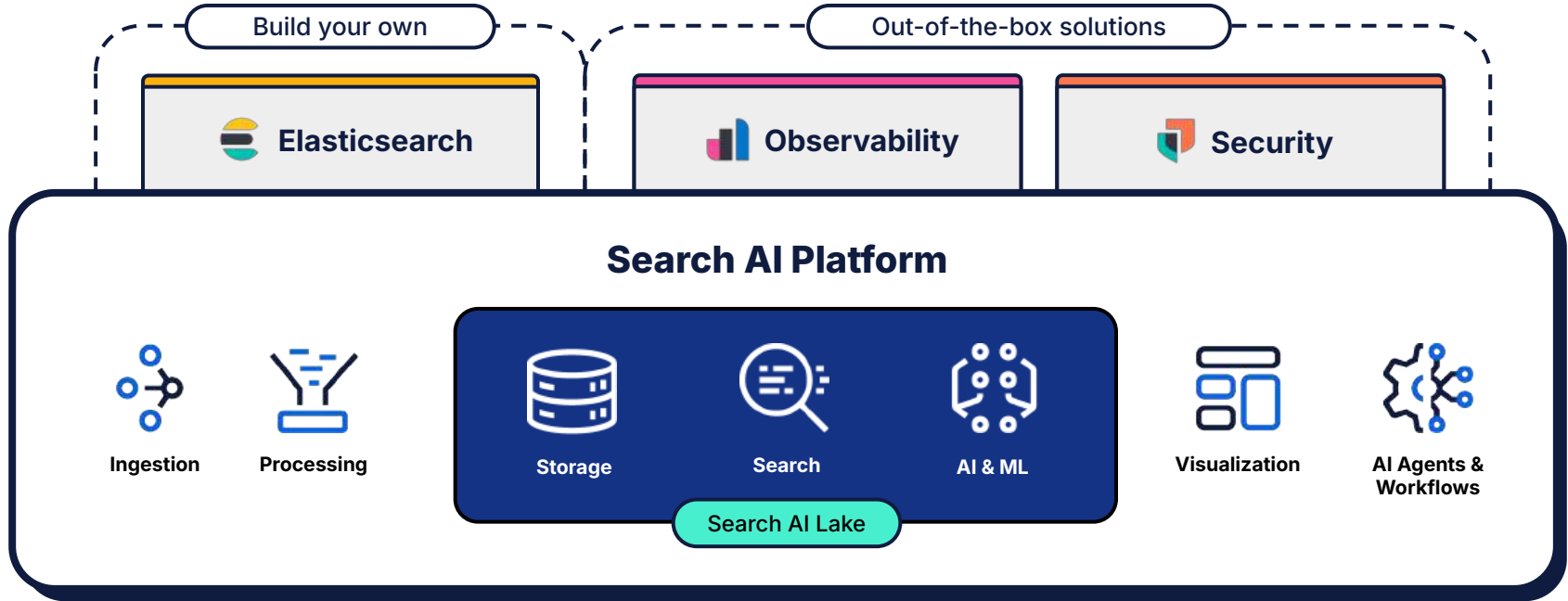
300+
Turnkey integrations

Meet Search AI

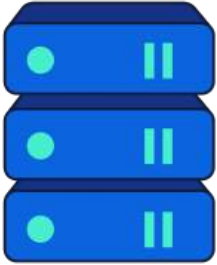
The fusion of **search technology** and **artificial intelligence**



One **platform**, two out-of-the-box **solutions**, the freedom to **build anything**



The Search AI Platform features a **whole new data architecture**



Data Warehouse

Response time:

10s of Seconds



Data Lake

Response time:

Minutes



Lakehouse

Response time:

Seconds



Search AI Lake

Response time:

Milliseconds

The Search AI Platform is available wherever your **data** lives

 Public cloud

 Hybrid

 On-premises

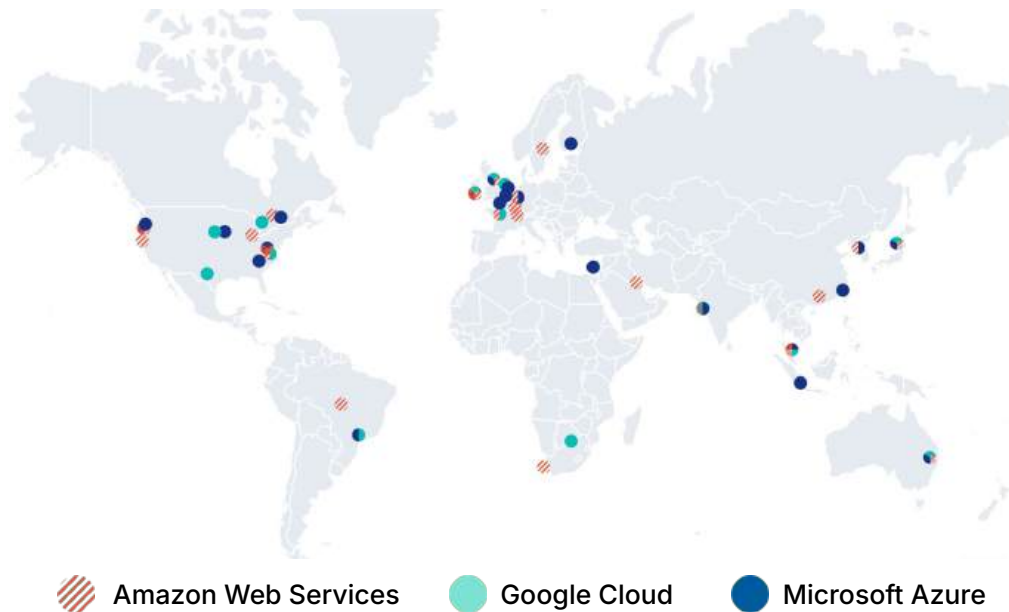


Google Cloud

Microsoft
Azure

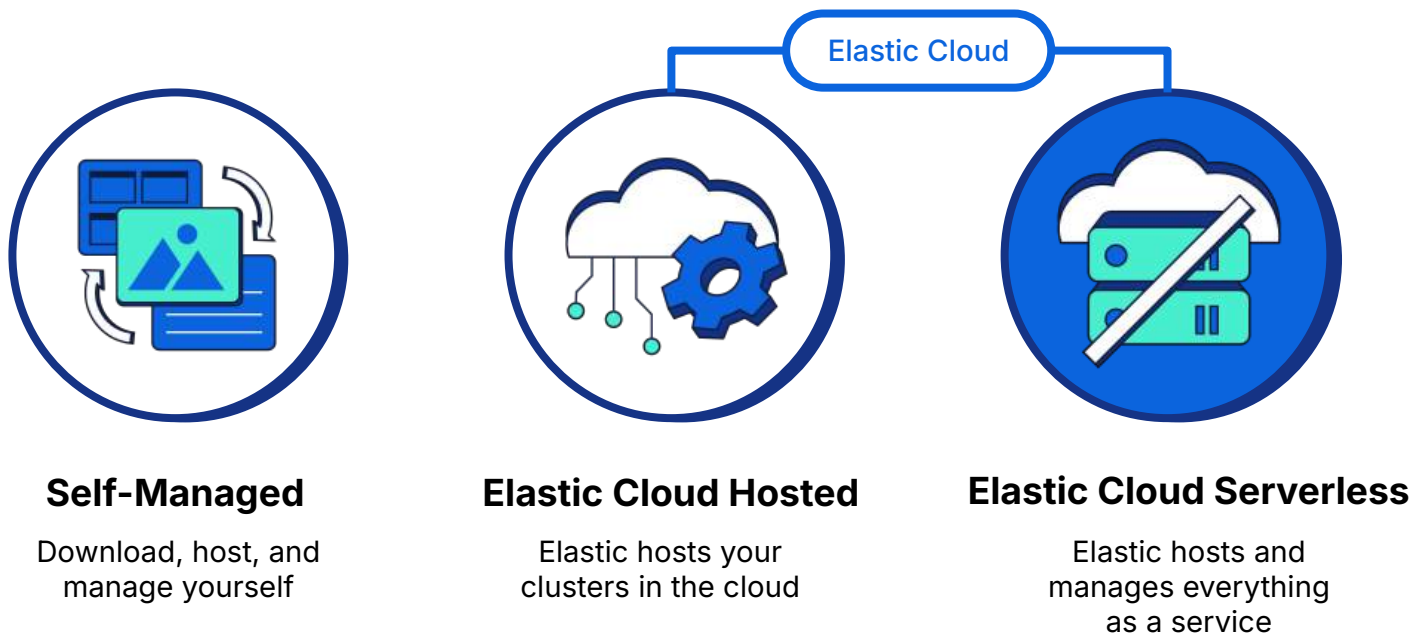

Alibaba Cloud


Tencent Cloud

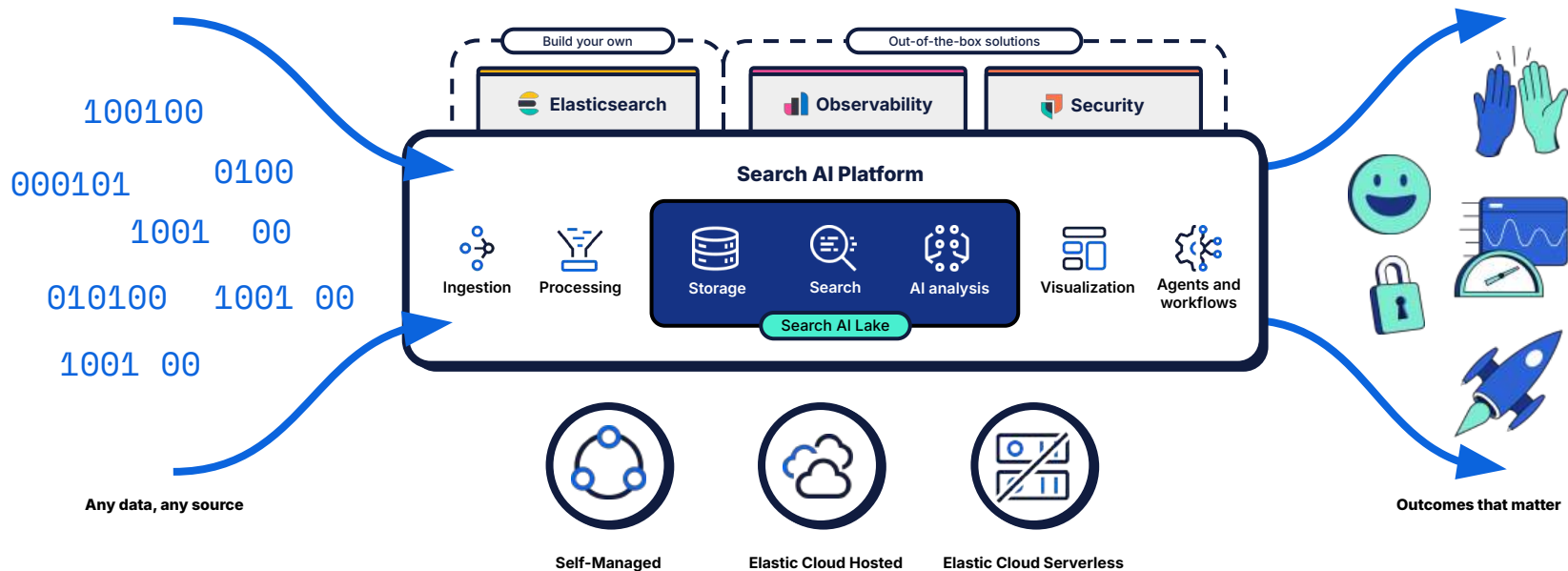


60+ cloud regions globally

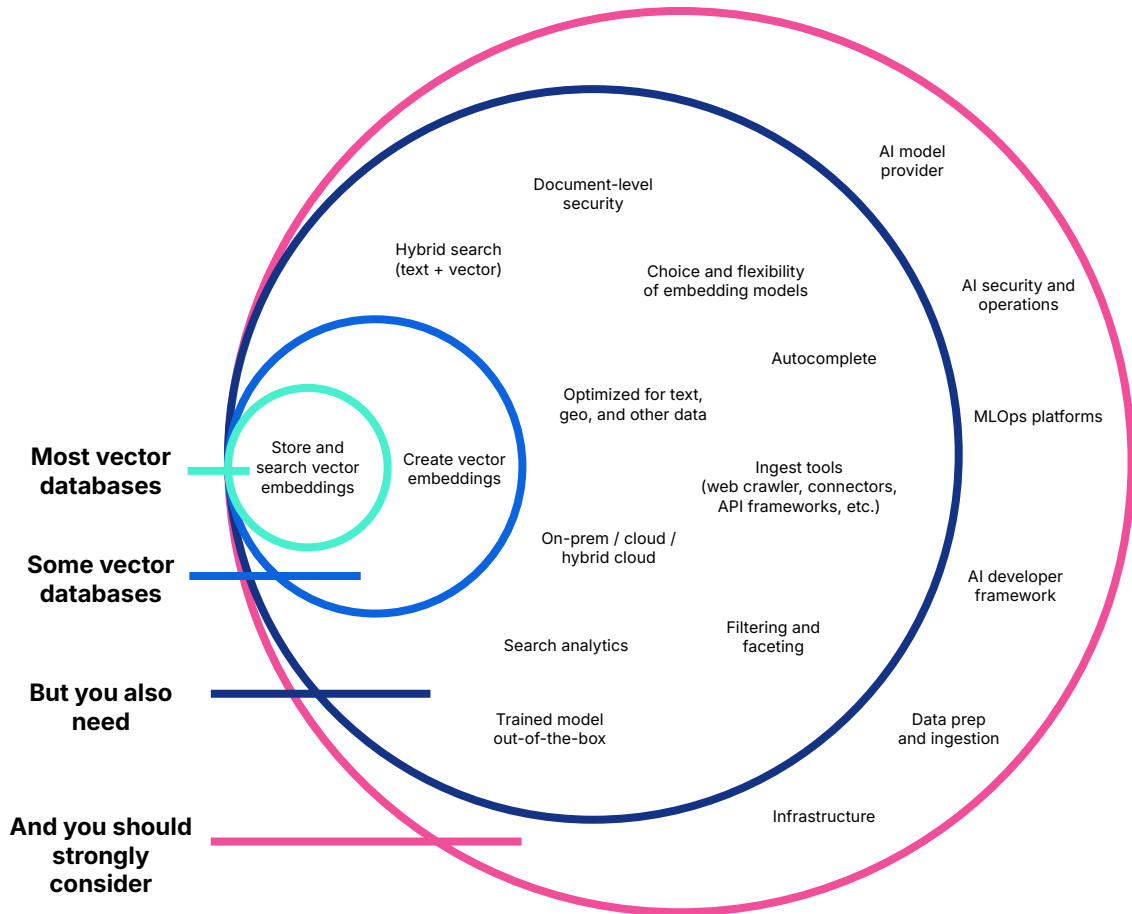
The **Search AI Platform** is deployable your way



Get 10x more value out of your **underutilized data** with **Search AI**



When it comes to building trusted, production-ready GenAI applications faster, **you need more** than just a vector database ...



Elastic AI-Powered Features

AI Assistant · Agent Builder · Workflows · Automatic Import · Streams

A focused guide to Elastic's five core AI capabilities — what they do, how to use them, and how they work together.

What We'll Cover

01

AI Assistant

Chat-based AI for queries, troubleshooting & threat investigation

02

Agent Builder

Build and deploy custom AI agents with tools, skills & connectors

03

Workflows

YAML-based automation with AI steps and bidirectional agent integration

04

Automatic Import

LLM-powered custom integration creation from any data sample

05

Streams

AI-assisted log field extraction, routing, and stream management

01

AI Assistant

Your conversational AI across Observability, Security & Search

AI Assistant — Overview

What is AI Assistant? (AI Agent in 9.4)

AI Assistant is a chat-based interactive tool embedded throughout Kibana. It helps users accomplish tasks using natural language — no query syntax required. It's available across all three Elastic solutions: Observability, Security, and Search.

Key Capabilities

- Construct queries (ES|QL, KQL, EQL) from natural language
- Call Elasticsearch APIs on your behalf
- Generate sample data for testing
- Create and explain Kibana visualizations
- Explain errors and suggest remediation steps
- Investigate threats and assist with alert triage
- Powered by Elastic Managed LLMs or your own (Azure, OpenAI, Bedrock)

Observability

Correlate logs/metrics, explain anomalies, query SLOs, assist root cause analysis

Security

Alert triage, threat research, investigate attacks, build detection rules

Search

Build semantic search, explain index mappings, optimize query performance

AI Assistant — Use Cases

Query Building

- Ask in plain English: 'Show me all errors in the last 24h'
- Converts to ES|QL, EQL, or KQL automatically
- Iterate conversationally — refine with follow-up questions

Incident Troubleshooting

- Paste an error message — get explanation + fix steps
- Ask: 'Why is my pod restarting?' — gets context from logs/metrics
- Auto-generates RCA summaries for postmortems

Security Alert Triage

- Analyzes alert context, enriches with threat intel
- Suggests MITRE ATT&CK mappings for detected behaviors
- Drafts case notes and escalation summaries

Dashboard & Visualization

- 'Create a dashboard for Kubernetes memory usage'
- Explains existing visualizations in plain language
- Suggests chart types based on your data shape

Data Onboarding Help

- Guides through index template and mapping creation
- Explains ILM policies and tier management
- Walks through ingest pipeline construction

Proactive AI Insights

- Log pattern analysis highlights anomalies automatically
- Surfaces related alerts and correlated signals
- Available in both Observability and Security

02

Agent Builder

Build, deploy, and orchestrate custom AI agents at scale

Agent Builder — Overview

What is Agent Builder?

Elastic Agent Builder is a platform for creating custom AI agents — LLMs equipped with tools, skills, and instructions that autonomously reason and act on user requests. Agents use an iterative tool-selection loop to achieve goals and return results.

Agents

LLMs with custom instructions + tools.
Built-in (Elastic AI Agent, Observability Agent, Threat Hunting Agent) or fully custom.

Tools

Actions an agent can perform: ES|QL queries, index search, MCP tools, workflow invocations.

Skills

Reusable instruction sets giving agents specialized expertise (e.g., log analysis, threat hunting).

Connectors

Wire agents to external services: Jira, PagerDuty, Slack, ServiceNow, and more via Kibana connectors.

MCP Server

Expose your Elastic agent externally via the Model Context Protocol for third-party AI integrations.

A2A Server

Agent-to-Agent protocol — allows other AI systems to call your Elastic agents programmatically.

Agent Builder — Use Cases

Observability Agent

- Ask: 'What's causing elevated error rate in checkout service?'
- Agent queries logs, metrics, and traces autonomously
- Returns correlated root cause with evidence chain

Threat Hunting Agent

- 'Find lateral movement in the last 48 hours'
- Runs ES|QL across security indices, cross-references threat intel
- Produces a structured threat summary with indicators

Custom ITSM Agent

- Create a Jira/ServiceNow-connected agent for ticketing
- Agent auto-creates, updates, and resolves incidents
- Can be triggered by alerts or conversations

Data Engineering Assistant

- Query Elasticsearch schemas, explain mappings
- Write and test ES|QL pipelines conversationally
- Assist with index lifecycle management decisions

Multi-Agent Orchestration

- Chain specialized agents: triage → enrich → respond
- A2A protocol enables agent-to-agent calls
- Combine Elastic agents with external AI systems

Embedded AI in Applications

- Expose agents via REST API or MCP server
- Embed intelligent search/Q&A in custom apps
- Build RAG applications on Elasticsearch data

AI Assistant vs Agent Builder

AI Assistant

VS

Agent Builder

Purpose	Guided assistant for Elastic tasks	Platform to build & run custom AI agents
Interaction	Chat in Kibana sidebar / flyout	Chat in standalone UI or embedded flyout
Knowledge Base	YES User-defined knowledge base	NO Not yet supported
Data Anonymization	YES Mask sensitive fields in prompts	NO Not yet supported
Custom Agents	NO One built-in assistant only	YES Unlimited custom agents
Custom Tools	NO Fixed toolset	YES ES QL, index search, MCP, workflow tools
Workflow Integration	NO Not supported	YES Call workflows; agents can trigger workflows
API Access	YES Observability & Security APIs	YES Full agent management & chat APIs
Chat Sharing	YES Share & duplicate conversations	NO Not yet supported
Best For	End users needing help with Elastic tasks	Engineers building AI-powered automation

03

Workflows

Deterministic, event-driven automation with built-in AI steps

Workflows — Overview

Elastic Workflows is a YAML-based automation engine inside Kibana. Workflows provide deterministic, auditable, repeatable automation — unlike AI agents, every workflow run follows the same logical path. They support AI steps that call LLMs or agents for reasoning within the workflow.

Triggers

- Manual · Scheduled (cron) · Alert-based (Kibana alerting rule fires a workflow)

Flow Steps

- If/Else conditions · Foreach loops · Wait (pause for time or human approval) · Switch routing

AI Steps

- ai.prompt (LLM call) · ai.classify (categorize data) · ai.summarize · ai.agent (invoke Agent Builder agent)

Action Steps

- Elasticsearch queries · Kibana API calls · External systems (Jira, Slack, PagerDuty, ServiceNow, webhooks)

Composition

- Child workflows · Reusable workflow libraries · Pass data between parent and child via output variables

Human-in-the-Loop

- Pause a workflow for analyst review before taking action — AI classifies, human approves, workflow proceeds

Workflows + Agent Builder — AI-Augmented Automation

Workflows and Agent Builder are complementary. Workflows provide deterministic, auditable automation. Agents provide reasoning. Together they enable automation that is both reliable and intelligent. The integration is bidirectional.

Workflow → Agent (ai.agent step)

Use when the workflow controls the flow and needs AI only for reasoning over a specific dataset.

Examples:

- Alert fires → fetch context → ai.agent summarizes → create Jira ticket
- ai.classify routes severity → different branches per level
- ai.summarize compiles evidence for analyst review
- Scheduled workflow enriches threat indicators nightly



Agent → Workflow (workflow tool)

Use when a user (or agent) is in a conversation and wants to trigger a deterministic procedure.

Examples:

- Chat: 'Isolate host webapp-03' → agent triggers isolation workflow
- Chat: 'Open a P2 ticket for this alert' → agent calls ticketing workflow
- Chat: 'Run enrichment on these IPs' → agent triggers enrichment workflow
- Conversational SOC playbook execution from chat

Workflows — Use Cases

Security: Alert Triage

- Alert fires → workflow enriches with threat intel
- AI step classifies severity → branches to response
- High severity: open case + page SOC; Low: log and close

Security: Detection Rule Management

- Scheduled workflow audits all detection rules
- Flags stale or duplicate rules for review
- Auto-disables rules with no hits in 30 days

Observability: Incident Response

- SLO breach alert triggers runbook workflow
- Gathers pod/node logs, recent deploys, error spike data
- ai.summarize compiles RCA for on-call engineer

Observability: Root Cause Analysis

- Alert on anomaly → workflow calls Observability Agent
- Agent correlates logs, metrics, traces autonomously
- Human-in-the-loop confirms RCA before paging

Data: Compliance Automation

- Scheduled daily audit of index retention policies
- ai.classify identifies data violating retention rules
- Sends summary report to compliance team via email

IT Ops: Auto-Remediation

- Disk/CPU alert triggers remediation workflow
- Tries known fixes (log rotation, cache flush) automatically
- Escalates to human if automated fix fails

04

Automatic Import

LLM-powered custom integration generation from any data sample

Automatic Import — Overview

Automatic Import uses LLMs to generate a complete Elastic integration from a sample of your raw data. It analyzes field structure, applies ECS normalization, generates categorization, and creates ingest pipelines — turning hours of manual work into minutes.

1

Upload Sample

Paste or upload a data sample
(JSON, NDJSON, CSV, syslog)

2

AI Analysis

LLM analyzes field types, names, and data patterns

3

ECS Mapping

Fields mapped to Elastic Common Schema automatically

4

Pipeline Gen

Ingest pipeline with Grok, dissect, & date processors

5

Review & Deploy

Inspect, edit, then deploy directly to your integration

400+ existing Elastic integrations available out of the box. Automatic Import is the path for custom/proprietary data sources not yet covered.

Automatic Import — Use Cases

Proprietary Application Logs

- Your app emits JSON logs with custom field names
- Automatic Import maps to ECS: @timestamp, host.name, etc.
- Ingest pipeline created and deployed in minutes

Network Device Syslogs

- Firewall/router syslog with custom format
- LLM generates Grok pattern to parse structured fields
- Immediately searchable in Kibana Discover

Third-Party SaaS Data

- SaaS vendor exports CSV/JSON not covered by existing integrations
- Automatic Import creates the mapping + pipeline automatically
- Normalized to ECS for cross-source correlation

Security SIEM Migration

- Migrating logs from another SIEM to Elastic
- Re-map field names from old schema to ECS in one step
- Detection rules can reference ECS fields immediately

IoT & OT Data

- Device telemetry with proprietary schemas
- Upload sample → auto-generate pipeline with type coercions
- Data ready for dashboards without manual coding

Accelerate Integration Development

- Start new integrations from AI-generated scaffold
- Engineers review and refine rather than build from scratch
- Reduce time-to-data from hours to minutes

05

Streams

AI-assisted log routing, field extraction, and stream management

Streams — Overview

Streams is Elastic's data routing and management layer for log data. It provides a UI-driven approach to partition data into logical streams, extract fields using AI-suggested Grok patterns, and manage data lifecycle — without writing pipelines by hand.

Stream Partitioning

- Route incoming logs into named streams by condition
- Separate streams for each service, environment, or team
- Each stream can have its own retention and ILM policy

AI Field Extraction

- Paste sample log lines — AI generates Grok pattern
- Preview parsed fields before committing to the stream
- No need to hand-write complex regex/Grok patterns

Stream Descriptions

- LLM-generated natural language description of each stream
- Auto-summarizes what data is in the stream and key fields
- Helps teams discover and understand data assets

Lifecycle Management

- Set hot/warm/frozen tier policies per stream
- Define retention duration per stream independently
- Reduces cost by right-sizing storage per data importance

Processing Rules

- Add field transformations and enrichments per stream
- Apply conditions to route sub-streams to different indices
- Works alongside ingest pipelines for maximum flexibility

Observability Integration

- Works natively with Elastic Observability log views
- Streams appear as data sources in Discover and dashboards
- Stream descriptions power AI-assisted search and exploration

Streams — Use Cases

Multi-Team Log Isolation

- Each team gets their own stream with separate access controls
- No cross-team data pollution or accidental overwrites
- Per-team dashboards with clean, isolated data views

Cost Optimization by Service

- High-value prod logs → hot tier with 90-day retention
- Dev/debug logs → frozen tier with 7-day retention
- Dramatically reduces storage costs vs. one-size-fits-all

Rapid Log Onboarding

- New service emits logs in unknown format
- AI generates Grok pattern from a paste of sample lines
- Stream live with field extraction in under 5 minutes

Security Log Segregation

- Route security-relevant logs to a dedicated stream
- Apply stricter retention and access controls
- Feeds directly into Elastic Security SIEM detection

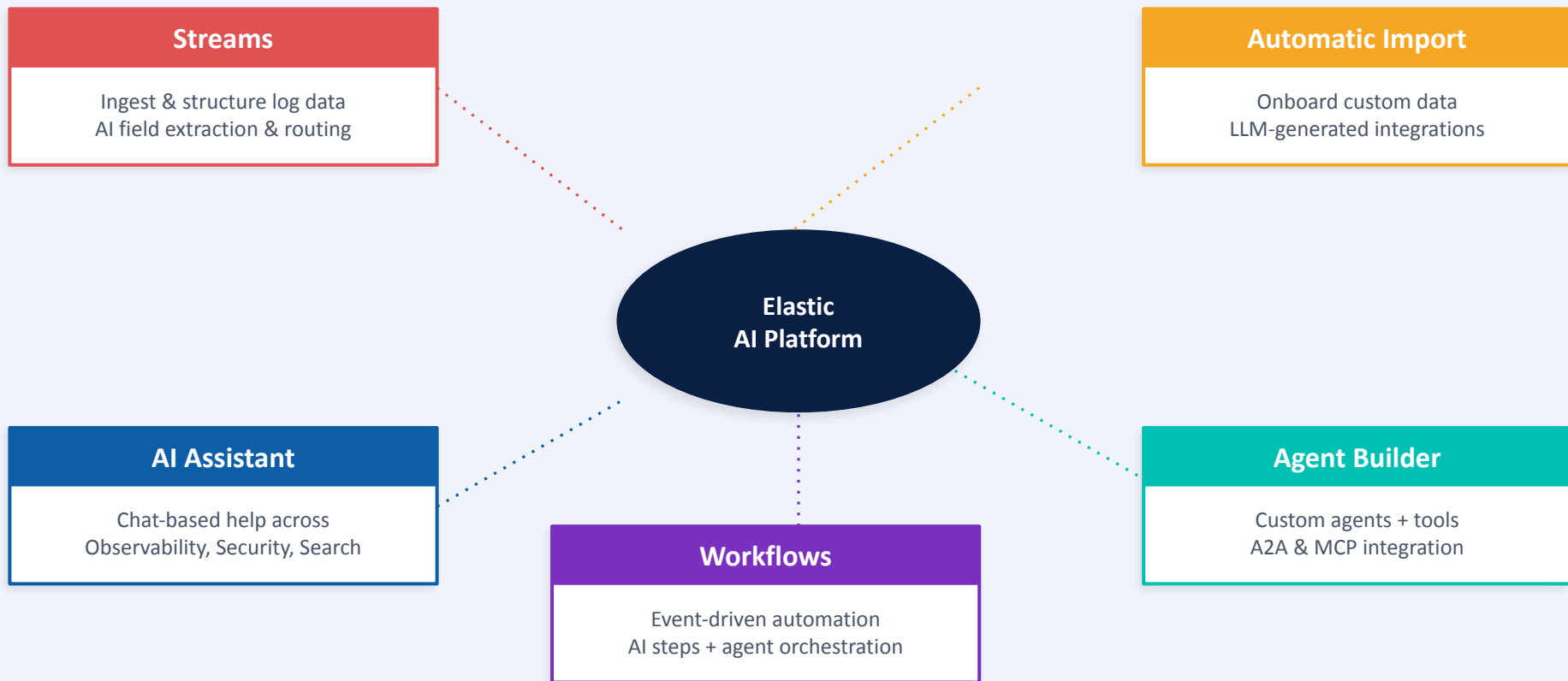
Environment Routing

- Route prod/staging/dev logs to separate streams automatically
- Prod stream: long retention + hot tier
- Dev stream: short retention + cheap storage

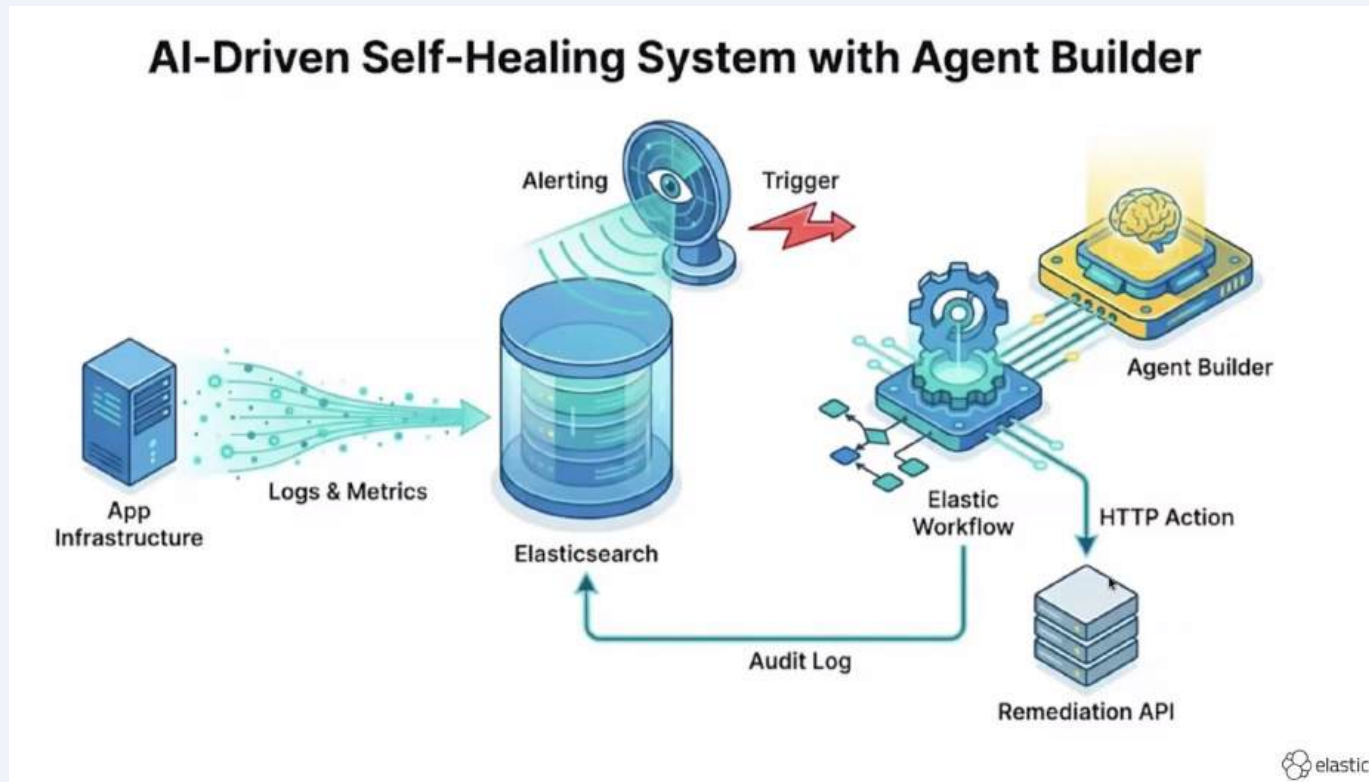
Data Discovery & Documentation

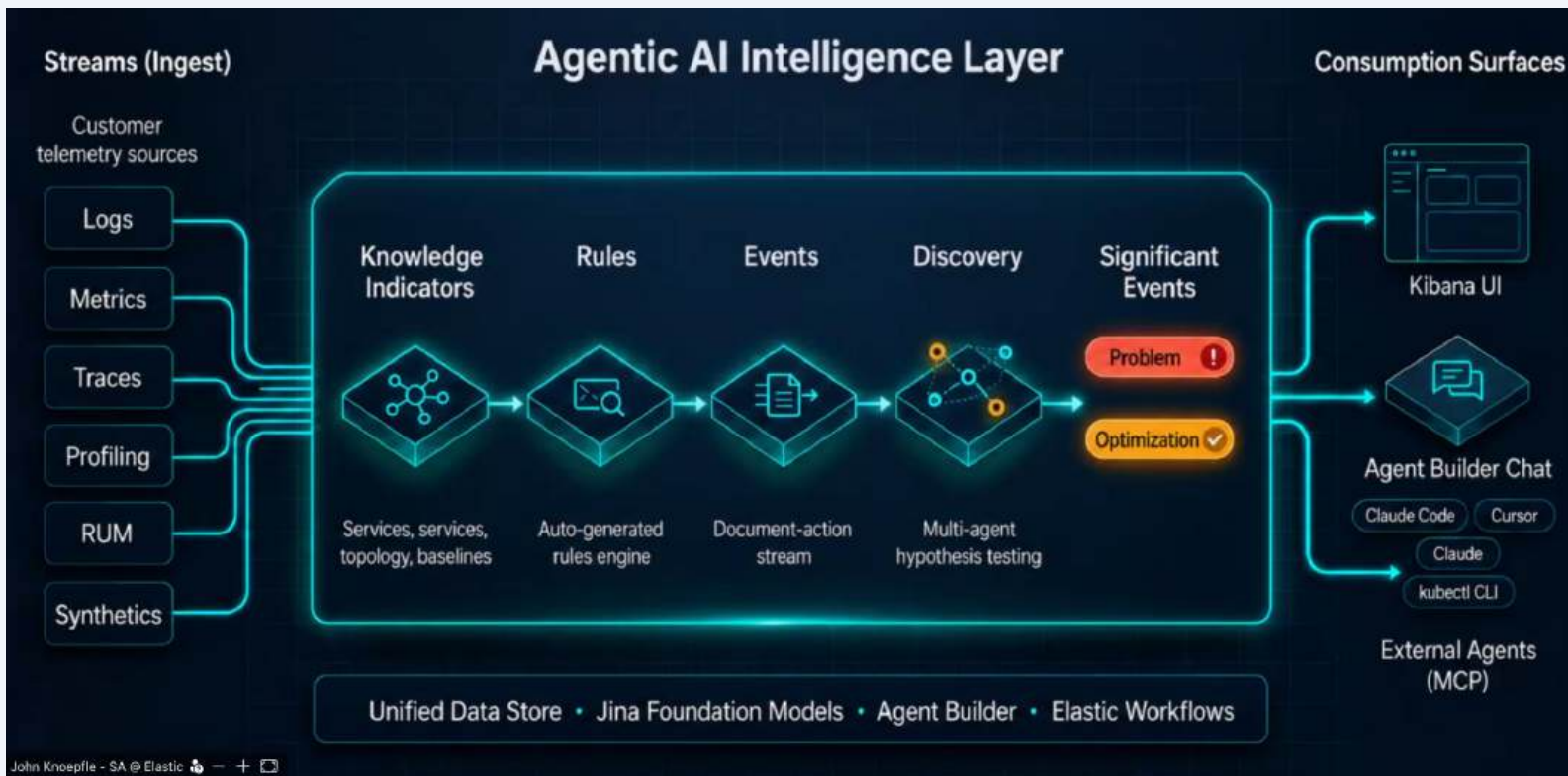
- AI-generated stream descriptions document what each stream contains
- New team members can search and understand data assets
- Reduces tribal knowledge dependency for data teams

How the Five Features Work Together



How the Five Features Work Together





Summary

AI Assistant

Natural language help embedded across Kibana — queries, troubleshooting, threat investigation, visualizations.

Agent Builder

Build custom AI agents with tools, skills, MCP, A2A — the platform powering autonomous AI operations in Elastic.

Workflows

Deterministic event-driven automation with AI steps — works bidirectionally with Agent Builder for AI-augmented SOC and ops playbooks.

Automatic Import

Onboard any custom data source in minutes — LLM generates ECS mapping, Grok patterns, and ingest pipelines from a data sample.

Streams

Route and structure log data with AI-assisted field extraction — manage retention, access, and lifecycle per stream.